EPHEMERIS

A generator of Berlin School music

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About

Ephemeris composes and plays long pieces in the manner of the modern Berlin School: sequencer lines of different lengths turning against each other over a common root, a bass that holds the ground, tape choirs and string machines, a lead that sings over the peak, the room of a tape echo and a long hall. Nothing is played back from a recording: every sound is synthesised by the engine -- the only recorded material are the single cycles some of the pad synth's wavetables are made of -- and every note is decided by the composer.

The same seed always gives the same piece. A piece is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

An ephemeris is the table that gives the positions of the heavenly bodies for every date. A sequencer is such a table, and the music here is made of orbits of different lengths around one root that meet again only now and then. The Rack page draws them that way.

Signal flow



The signal flow: every unit a box in the colour of its family, the groups as frames, the buses as arrows.

What plays the instruments comes in at the top: the composer, who writes the form, the harmony, the sounds and the mix of every piece; the rack, eight sequencer rows and the transposer that moves them; the moves, the automation of a player's two hands on the knobs; and you, on the Perform page or with MIDI. The ten modular voices -- the rows, the lead and the drone -- run side by side in the lanes of the processor's vector registers at twice the sample rate: two oscillators after one of six classic designs, a driven mixer, one of ten filters solved as circuits, envelopes, LFOs and a matrix. The other sources play beside them. Every source has a channel strip that places it in depth and in the stereo field and sends it into the rooms; the master bus keeps the lows mono, the width in check and the peaks under -1 dBTP, and a correction measured on the piece's loudest part brings every piece to its style's level.

The top bar

Style, key and scale choose the piece's character: Cosmic, Doom, Melodic, Modern and Drift are five style profiles with their own tempi, forms, layers and spaces. Beside them, three buttons choose what is composed: Piece -- a single piece; Concert -- pieces strung together into a concert that travels through related keys; Night set -- pieces in mixed styles, each mixed into the next as a DJ does it. Length is the length of what is chosen: a piece's 4 to 40 minutes, a concert's or a night set's 20 minutes to 12 hours. A button composes what it names at once if what plays is something else; Compose names what it makes (Compose piece, Compose concert, Compose night set) and is lit while what plays is not what the buttons choose. The status row says what plays: a single piece, or a concert or a night set of so many pieces. With Morph To (on the Master page) a concert also wanders from its style to another: each piece is composed between the two, the further into the concert the nearer the second. Concert Arc gives a concert an arc of tension: its first and last pieces quieter -- fewer rows at the peak, the lead and the drums rarer, the moves slower, the sound a little darker -- and the pieces around 60 % of the way the densest. Compose writes it with the current seed, the die draws another one, Play (the triangle, a square while it plays) starts and stops the standalone (in a host the host's transport plays), the speaker silences it. Undo, Redo, Help and the settings are at the right of the row below (The panel).

A night set (up to twelve hours) is played as a DJ plays one. The pieces last eight to fifteen minutes and their styles climb and descend a ladder -- Drift, Doom, Cosmic, Modern, Melodic -- one rung at a time, along waves of energy of 80 to 120 minutes over the night's own arc (the densest around 60 % of the way), never three pieces of one style in a row; the style in the top bar is the first. No piece ends in silence: each one's last 32 to 40 bars are an outro in which its rows play on, and the next piece starts over it, on the other half of the rack (rows 1-4 and 5-8 take turns). First its intro, 8 to 16 bars -- the drone slides into the new key, a fourth or a fifth away (now and then a minor third), with its pads and its air -- then its bass, alone for eight bars: there the old piece's counter rows stop and its bass gives up its lows on the EQ and recedes, as a DJ swaps the bass. Then the new main sequence comes in and fills up while the old one recedes to its end. The mix is beat-matched: the new piece takes the tempo the old one ended in, within 6 BPM of its style's range, and ramps to its own only after the handover (16 bars for every 4 BPM). The level stays the old piece's until its end, and Doom and Drift are brought 1.2 dB nearer to the louder styles. A piece of a night set has at most four rows. Composing eight hours takes about half a second; export writes the whole night.

Scale offers eight modes: Aeolian, Dorian, Phrygian, Harmonic Minor, Minor Pentatonic, Mixolydian, Lydian and Locrian. The harmony is the Berlin School's: a centre held by the drone and the bass, chords that change rarely. From the first build on, a chord track moves the bass row alone under the unchanged sequence -- the counter rows are heard anew over the wandering bass -- in one of four ways the style draws: one chord for the whole phase, a pendulum between two (i-VII, i-VI, i-iv, i-IV in Dorian, i-bII in Phrygian), a loop of three or four (i-VI-VII, i-VII-VI, i-iv-VI ...) or a walk by a table of chances. Drift and Modern stay longest on one chord, Melodic changes every four to eight bars. The tape keys and the strings build their chords on the same degree: the choirs mostly plain triads, the strings open fifths, triads, sus chords, add9, stacked fourths and m9 as well. The transposer moves the whole sequence a fifth up, a fourth up or a minor third down and back, only on the plateau and at the peak, while the drone stays on the centre. At some peaks Aeolian brightens to Dorian, in some breakdowns it darkens to Phrygian, and every piece ends on the open fifth. A later phase may move its key by a fourth, a tone or a third, and take a minor mode of its own, as the parts of a suite do.

The reroll buttons (on the Arrange page) draw one part of the piece again and leave everything else as it was, to the note: the form, the tempo, the rows' shapes, the rack's patterns, the layers, the lead, the chords and pads, the moves on the knobs, and the sounds. The disk and the folder on the Export page save and load a piece as a small text file; Export writes the piece offline as a 24-bit WAV and a MIDI file beside it, with the moves as controllers -- and, if you choose so, the stems: a WAV per channel strip and one for the rooms in a folder beside the mix, whose sum is the mix before the master.

The panel. Every synth's page is laid out as an instrument's panel: its controls in titled groups -- Oscillators, Wavetable, Filter, the Amp, Filter and Mod Envelopes, the LFOs and the Mod Matrix, Mix, Sends, and on the string machine Registers, Tone, Envelope and Ensemble as on a Streichfett -- and the controls a hand goes to first as large encoders: the cutoff and the resonance, the registration and the ensemble, the pad synth's place in its table, the level. The colours say what a control does, the same on every page -- they are the logo's five planets: amber for the sources (oscillators, tapes, registers), copper for the filters, sage for the envelopes, teal for everything that moves by itself (LFOs, scans, ensembles, the echoes' time), steel blue for the room and the mix; the sun's amber marks the page in front and the transport. A page taller than the window scrolls. The standalone's window can be maximised, and full screen (F11, or in the settings; Esc leaves it) fills the screen; the whole panel grows with the window. The update check (in the settings) asks GitHub once a day whether a newer Ephemeris is out and shows it in the status row as a link to its page -- nothing else is sent, nothing is downloaded; switched off, it asks nothing. The synths that share a tab -- the row voices, the lead, the drone, the poly synth, the tape keys and the strings under Synths, the echoes and the hall under Rooms, the console and the master under Mixer -- each have a small tab of their own. A jump on the arrange view finds the notes that sound on across it -- the drone, the pads, a long lead note -- and starts them where it lands.

Below, the arrange view shows the sections of the piece -- the atmosphere at the start, the entry of the rows, the builds, the lead, the peak, the bridges, the fade -- with the playhead, a lane per layer lit where it has notes, and a ruler of bars (of minutes for a concert or a night set). Click to jump. A concert's pieces are apart by a line and carry their number (a night set's its style as well).



A concert of four pieces in the arrange view: their numbers at their starts, the ruler in minutes.

The mouse wheel zooms in and out around the pointer, down to four bars; zoomed in, the lanes show the notes themselves, a bar along the bottom shows where the window lies in the whole, and the window pages on with the playhead -- unless it was moved away from it. A drag, Shift with the wheel or the wheel sideways moves along; a double click shows the whole length again.



The arrange view zoomed in on forty bars of a build: the rows' notes, the pads, the drone, the bar numbers.

The panel

Ephemeris shares its panel with the other generators of the family -- Phosphene, Totality and Parhelion (and, in part, Noctuary) -- so that what is learned on one holds on the others: the same places, the same keys, the same controllers, the same hands. Each keeps its own colours, its own letters and the picture behind its panel (Settings > Picture behind the panel takes it away).

The header's first row: the logo, the style, the key and the scale, what is composed and its length, Compose, the die (a new seed), Play and the speaker (mute). Its second row: the status line, and at the right Undo, Redo, Help and the settings. Under the header the whole piece, concert or night set at a glance; a click jumps there. The tabs come in one order everywhere: Set, Arrange, the instrument's own pages -- a page of several has small tabs of its own --, Mixer, Perform, Export, Style.

The unambiguous actions are icons since 03.10.2026, their words in the tooltips: the die draws a new seed, the triangle plays and the square stops, the speaker mutes (crossed out while muted), the disk saves and the folder loads, the MIDI socket learns a controller (Perform).

The ways around the panel (15.10.2026, the family's). The magnifier at the right of the header (Ctrl+K) opens the search: a few letters find any knob -- its page opens, its section or lane comes to the front and it lights up; a number after its name sets it ("cutoff 800") --, any preset of any synth, a page or an action. The line along the bottom of the window says what the control under the mouse is: its value now, its range and default, where the score plays it, what moves it, its MIDI controller and its key. Beside a synth's presets on its page sits a magnifier, the browser: by name, by group, the starred and the recently played; hold a row's triangle to hear it, double-click to take it. The disk beside saves the sound as a preset of your own, listed under Yours, and a star says that the knobs have moved from the preset. A knob a voice's modulation moves wears an arc round it for every row of the matrix on it -- the span the row can move it over, in its source's colour; a right click lists those rows, each with its depth to drag and a button to take it out, and offers every source (Modulate with: a row in the first free slot, a quarter of the target's span); an Alt-drag on such a knob sets its last row's depth (up deeper, down the other way). The first steps show the panel on the first start (Settings > First steps, again). The standalone's window comes back where it was left -- its place, its size, maximised or not --, and its audio settings offer ASIO (Windows).

A page taller than the window shows its groups in sections, one at a time, with a switch at its top right: a synth's sound and its modulation apart (Sound, Modulation), and where even those do not fit, cut further, each named after its first group (Filter, LFO, Matrix ...). At the window's usual size, 1280 by 860, no page scrolls. The overview above the tabs can be folded away in the settings; the pages then get its height.

A colour says what a control does, the same on every page and in every generator of the family: the sources, the filters, the envelopes, what moves by itself, the room and the mix. A knob the composer moves has a thin ring round it that shows where it stands at this moment. A double click sets a control to its default; a right click offers MIDI learn, forgetting the controller, and the default.

Undo and Redo take back and bring back what was changed -- a knob, a new seed, a reroll, a preset, a loaded set -- one step at a time; the button's tooltip names the step. Help opens this manual by topic, with the page in front, the keys and the headset as topics of their own. The settings hold the update check (it asks GitHub once a day whether a newer Ephemeris is out; nothing else is sent), the picture behind the panel, the overview, the headset, the window's size, full screen, the keys and About.

```
Space                 play / stop
Ctrl+Z                undo
Ctrl+Y, Ctrl+Shift+Z  redo
F1                    help
F11                   full screen (the standalone); Esc leaves it and closes the help
Ctrl+S, Ctrl+O        save the set, open one
Ctrl+E                export
Ctrl+K                search: any control, preset, page or action
```

The controllers every generator of the family listens to without learning: controller 74 (brightness) moves the rows' filters, 11 (the expression pedal) throws the echo, 64 (the sustain pedal) holds the moves; the mod wheel is left free.

The headset. The controls of a Meta Quest's hands are shown only while a headset sends them -- or always, if the settings say so (Headset > Always show them); Off stops listening. The Quest app sends its hands to this computer in bridge mode: bridge_host = this computer's address in its eph.cfg (audio=0 leaves the headset silent), and Ephemeris listens on port 9102. The hands mean the same in every generator of the family, on the headset alone and here:

```
left pinch            play / stop
both hands pinched    the next piece (a new seed)
right pinch           hold the moves, or let them go again (Hold Moves)
left hand up, down    the rows' filters, two octaves either way
right hand up         the echo throw
```

A pinch acts when it opens again, so a pinch of both hands is never also one of each; a hand moves its control only while it does not pinch, and both glide and are centred: a hand at mid height plays what was composed.

Set



The composer's settings: the style, the key, the scale, the tempo and whether the style chooses it, the lengths of a piece, a concert and a night set, Morph To, Concert Arc and Album Form -- the top bar shows the first of them as well. Their meaning is told under The top bar and Master.

COMPOSE

Control	Key	Range	Default
Tempo	`compose.bpm`	60 .. 160 BPM	118 BPM
Key	`compose.key`	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A
Scale	`compose.scale`	Aeolian, Dorian, Phrygian, Harmonic Minor, Minor Pentatonic, Mixolydian, Lydian, Locrian	Aeolian
Style	`compose.style`	Cosmic, Doom, Melodic, Modern, Drift	Cosmic
Piece Length	`compose.piece_minutes`	4 .. 40 min	16 min
Style Tempo	`compose.style_tempo`	off / on	on
Concert Length	`compose.concert_minutes`	0 .. 720 min	0 min
Morph To	`compose.morph_to`	None, Cosmic, Doom, Melodic, Modern, Drift	None
Concert Arc	`compose.concert_arc`	0 .. 1	0
Album Form	`compose.album`	off / on	off
Night Set	`compose.night_set`	off / on	off
Composer's Sounds	`compose.pick_sounds`	off / on	on

Mixer



A channel strip per source, in the order the engine mixes them: the eight rows, the lead, the drone, the tape keys, the string machine, the drums and the atmosphere. Each strip has its pan (where the source has one), its sends to the tape echo and to the hall -- the rows also to the second echo -- its fader and its meter. The controls are the same parameters the sources' own pages show. Show sends unfolds the sends of every strip (the echoes, the early reflections, the blend room, the hall, the shimmer); folded, the console shows level, pan and meter only, with long faders.

The composer sets the stereo field for each piece: the bass in the middle, the main sequence near it, the other counter rows 30 to 50 % out to alternating sides, the lead a little off the middle. It keeps the bass dry and the tape keys and strings out of the echo, and sends the counter rows after the first into the second echo, so each layer of the sequence has a delay time of its own. It sets all this on the faders, pans and sends themselves as the piece begins -- the mixer shows the piece's mix, and a hand that moves a fader moves it from there; its moves along the form (the bass coming closer, the pads stepping back while the lead plays) run from those settings. As with the sounds, a new piece that keeps a strip's settings keeps your changes to it.

Every source has a Low Cut on its own page: a gentle high pass (12 dB per octave) that keeps it out of the bands below its own, after the production guide's frequency plan. Only the bass row reaches down to 30 Hz; the composer starts the main sequence at 90 Hz and the other counter rows at 200 Hz; the lead and the tape keys start at 150 Hz, the strings at 200 Hz, the atmosphere at 120 Hz, the drone at 40 Hz (its lowest note is at 82 Hz), the drums at 25 Hz. Mud between 200 and 500 Hz and a crowded bass are what make a long piece tiring; this is where they are kept away.

Punch (on the rows) lifts a row's attacks by up to 6 dB and leaves its sustain: the composer gives it to the main sequence, which keeps it in front. Distance (on the rows, the lead, the drone, the tape keys and the strings) places a source between near (0) and the horizon (1) with every cue at once, so nothing is near in its level and far in the hall: at 0.3 it is 4 dB quieter, its highs end at 10 kHz and it sends a little more to the hall; at 0.6 it is 10 dB back, dark from 5 kHz and well in the hall; at 1 it is 24 dB back, 2.5 kHz and almost only hall. The composer moves the bass with it: at each entry it comes in from 0.7 over 45 seconds, in the breakdown it recedes over up to two minutes, and in the coda before it stops -- an approach and a leaving instead of a fader ride.

The rows also make room for themselves: while they play, the pads (strings and tape keys) step back in their middle, by up to Cascade Duck (Master page, 2.5 dB) -- in six bands between 350 Hz and 4.8 kHz, each only where the rows have energy in it -- and the pads in turn push the atmosphere back the same way: the pads stay full while the sequence rests. The rows reach the mix through a resonance

suppressor (Resonance Tamer, Master page): in the same six bands, a band that stands more than 6 dB over the others is pulled back, up to 3:1, so a filter's resonance sweeping through never sticks out or tires the ear.

The meter reads what the source puts into the mix after its fader and pan: a green bar for the RMS level with a release of 300 ms, turning amber above -6 dBFS and red above 0, and a peak line that holds for a second and a half before it falls at 20 dB per second. The figure under the meter is the held peak. A strip that shows -inf is not playing at the moment.

Perform



The performer's controls act on the piece as it plays, and at their defaults they do nothing at all: the piece is then exactly as composed. Filter moves the rows' filters up to two octaves either way, as a hand on the cutoff knobs of a modular would. Transpose is the transposition key: every note that starts after a change comes in transposed; notes that sound keep their pitch. Hold Moves lets go of the knobs: the moves the composer wrote stand where they are until you release it. Echo Throw sends every source into the tape echo and raises the echo's feedback with it, for a dub throw at the end of a phrase.

A MIDI keyboard plays the transposition key directly: a key transposes by its distance from middle C and holds until the next key, as the transposer of a sequencer did. Controller 74 (brightness) grabs the filters, the expression pedal (11) throws the echo, the sustain pedal (64) holds the moves -- the same controllers in every generator of the family. The MIDI socket beside each binds any controller to it; the bindings are saved with the plugin's state. While a headset sends its hands, a box beside the controls shows what they do (The panel).

The keyboard (the Keyboard group): Keyboard Plays sends the keys of a MIDI keyboard to a voice instead of the transposition -- the lead, the drone, the poly synth, the tape keys, the strings, the drums (General MIDI's drum map), a row, or by channel (1 to 8 the rows, 9 the lead, 10 the drums, 11 the drone, 12 the poly synth, 13 the tape keys, 14 the strings) -- with the sound its page has, untransposed. Replace leaves that voice's generated notes out, Layer plays over them; Composer off leaves every generated note out, and only what is played sounds, through the mix and the effects as composed. The keys play while the music runs, each on its sample; an export plays what was composed.

Three options of the Keyboard group, each off until you choose it: Lower Keys Play splits the keyboard -- the keys below Split At (C1 to C5, C3 to begin with) play the voice chosen there, the keys from it upwards the voice of Keyboard Plays; Scale Lock moves a key outside the piece's scale to the nearest note inside it, so no wrong note can sound; Velocity Curve shapes the touch -- As Played, Soft (a light touch is enough), Hard (it takes a firm one) or Fixed (every note at 100, for pads that send no velocity). A key's release always ends the note its press started, even when an option changed while it was held.

A row cannot be mutated live: the piece is composed before it plays. To hear the rack with other patterns, reroll the rack.

PERFORM

Control	Key	Range	Default
Filter	`perform.filter`	-2 .. 2 oct	0 oct
Transpose	`perform.transpose`	-12 .. 12 st	0 st
Hold Moves	`perform.hold`	off / on	off
Echo Throw	`perform.throw`	0 .. 1	0
Keyboard Plays	`perform.keyboard_part`	Off, Lead, Drone, Poly, Tape Keys, Strings, Drums, Row 1, Row 2, Row 3, Row 4, Row 5, Row 6, Row 7, Row 8, By channel	Off
Keyboard Mode	`perform.keyboard_mode`	Replace, Layer	Replace
Composer	`perform.composer`	off / on	on
Lower Keys Play	`perform.keyboard_lower`	Off, Lead, Drone, Poly, Tape Keys, Strings, Drums, Row 1, Row 2, Row 3, Row 4, Row 5, Row 6, Row 7, Row 8	Off
Split At	`perform.keyboard_split`	C1, C2, C3, C4, C5	C3
Scale Lock	`perform.keyboard_scale`	off / on	off
Velocity Curve	`perform.keyboard_velocity`	As Played, Soft, Hard, Fixed	As Played

Rack



On the left, the orrery. Every row that plays notes is a planet on an orbit, the bass innermost; one turn is one cycle of the row, its length in steps times its step. The top of every orbit is the row's first step. When running rows stand there together, a line of light rises from the centre: a conjunction, the moment the patterns come round together. Under the orbits, each row's cycle and the time until all running rows meet again. The sun in the middle is the transposer: its letter is the root the rows play on at the moment, and it moves when the transposer does.

The composer uses the conjunctions: a build, which brings in a new counter row, ends where that row and the bass row begin their cycles together again, if such a bar is near.

On the right, the rows' parameters: length and step, direction, octave and transposition, the chance of a mutation (a Turing machine's shift register) -- weighed once every sixteen bars, so that a pattern holds at least that long, as a hypnotic one must -- the gate, level, pan and sends, the filter sweep, and the mode -- notes, or the transposer that moves the others. The composer sets the rows for each piece; the values here are what a piece starts from.

The composer draws each row from one of the Berlin School's figures: the octave pendulum, the fifth anchor, the 3+1 pulse (three notes and a rest), stairs up and down, the mode's colour (the note that makes Dorian or Phrygian heard), the Phrygian push, the arpeggio spiral from i to VI, or a canon of the row before -- three steps later or a fifth up -- besides the older free rules. The bass ostinato has eight or sixteen steps and a long gate; every eight steps have one to three rests (the bass none to two), the first always sounds on the root or the fifth, accents fall on the first and fifth step or 3+3+2. The bass and the main sequence hold the ground: they change at half the style's rate and never by chance. On the other counter rows some steps are probability gates that sound on some

rounds only -- Modern has the most, Cosmic and Doom the fewest; a few steps, the most in Drift and Modern, play a new note each time round, drawn through a quantiser from the root, fifth, octave, seventh, fourth and third of the mode. A counter row comes in with six of its steps silent and fills a step every two bars. On the plateau and at the peak a few steps are ratcheted into quick triggers, at some peaks the eighths row doubles its pulse, and in the breakdown and the coda the bass loses its steps one by one.

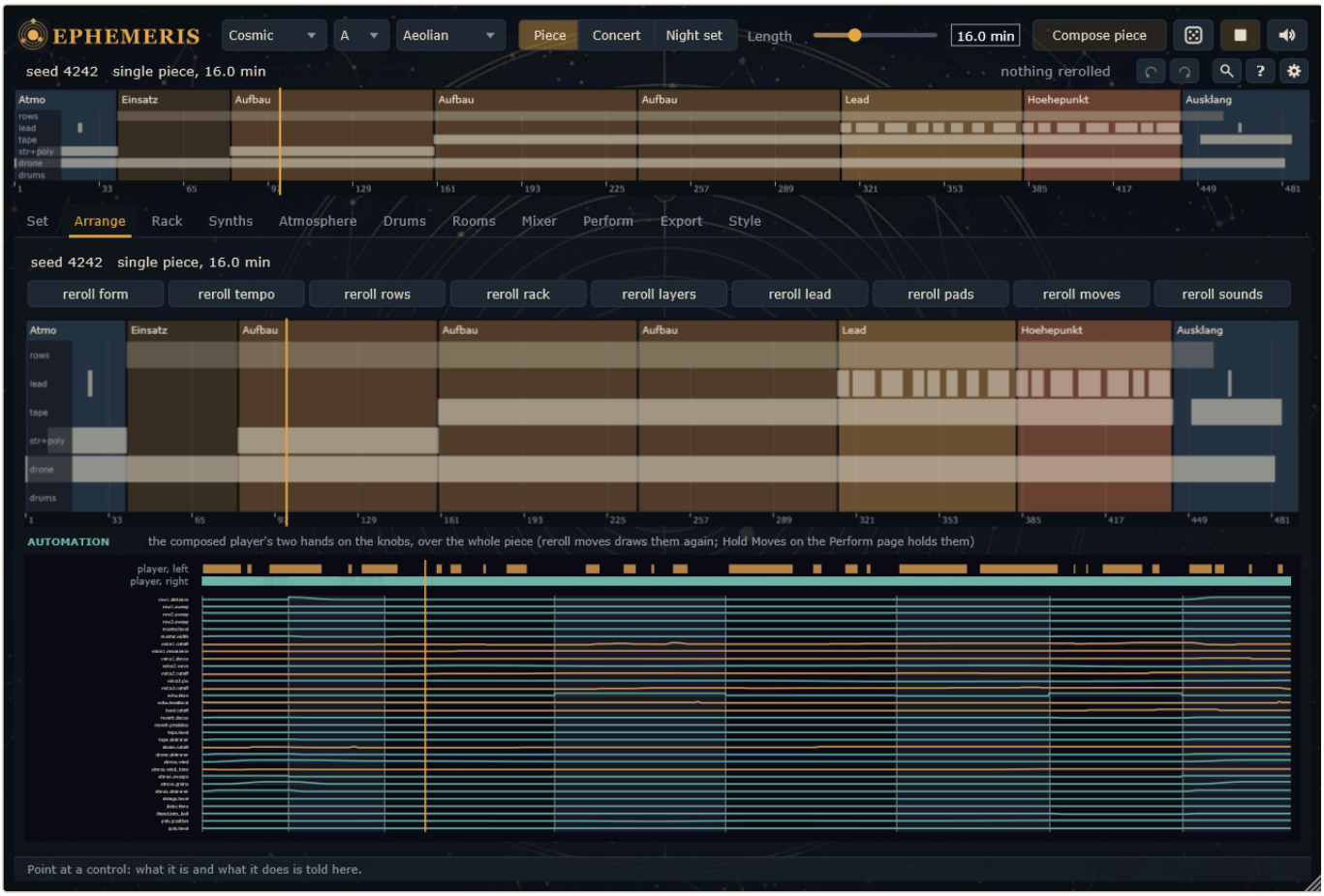
What makes a sequence hypnotic is less its notes than its timbre moving against them. Every row carries two modulation sequencers beside its notes: a lane of 3, 5, 7 or 12 steps -- never the row's own length -- that opens the filter on a few notes and closes it on others, and a second one of 3 to 12 steps that lengthens the filter's decay on some notes and plucks others short. The accents of timbre wander through the pattern and meet it again only after many rounds (Modern the deepest, Drift the shallowest). Over that, Filter Sweep turns the row's cutoff slowly up and down, a sine of its own period between 39 and 79 seconds for every row, never in step with the bars or with each other: the long openings and closings of the Berlin School. The composer gives the main sequence the widest sweep and a squelchy voice -- more resonance, a short deep filter envelope, strong accents -- the bass a round one and the other rows something between; Motion on the Master page scales every sweep with the other slow movements. And on every counter row one knob of its timbre wanders through the piece in glides of one to three minutes -- the place in its wavetable, else the mix of saw and pulse, or the pulse width where the pulse is heard -- as does the pad synth's place in its table: a spectrum that never stands still for long. The bass keeps its sound, the ground.

No sequence runs more than sixteen bars unchanged. The composer plans the variations the way a player at the sequencer makes them, one at a time, every eight to sixteen bars: a step switched off or on, a step thrown an octave up, the row shortened by a step or two for four or eight bars (the sequencer skipping), its direction turned backwards or into a pendulum for as long, or the row taken back to its theme once two things have changed -- mostly on the main sequence, now and then on another counter row or the bass. Before some peaks there is a break: every row and the drums stop for two or four bars, the echoes' tails, the pads and the lead go on with the echo's feedback raised, and everything comes back together on the peak's downbeat (Melodic and Modern in three pieces of five, Drift in one). One to three times a piece the tape echo switches from its dotted eighths to eighths or quarter triplets for a section.

ROW (8 INSTANCES)

Control	Key	Range	Default
Active	`rowN.active`	off / on	on
Length	`rowN.length`	1 .. 32 steps	16 steps
Division	`rowN.division`	1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1 Bar, 2 Bars, 4 Bars	1/16
Direction	`rowN.direction`	Forward, Backward, Pendulum, Random Walk	Forward
Octave	`rowN.octave`	-3 .. 3	-1
Transpose	`rowN.transpose`	-12 .. 12 st	0 st
Mutation	`rowN.mutation`	0 .. 1	0
Gate	`rowN.gate`	5 .. 100 %	50 %
Level	`rowN.level`	-60 .. 6 dB	-6 dB
Pan	`rowN.pan`	-1 .. 1	0
Echo Send	`rowN.echo`	0 .. 1	0.3
Mode	`rowN.mode`	Notes, Transposer	Notes
Reverb Send	`rowN.reverb`	0 .. 1	0.3
Echo 2 Send	`rowN.echo2`	0 .. 1	0
Low Cut	`rowN.low_cut`	10 .. 500 Hz	30 Hz
Distance	`rowN.distance`	0 .. 1	0
Blend Send	`rowN.blend`	0 .. 1	0
Punch	`rowN.punch`	0 .. 1	0
Early Send	`rowN.early`	0 .. 1	0
Filter Sweep	`rowN.sweep`	0 .. 2 oct	0 oct

Arrange and automation



The Arrange page: the rerolls, which draw one part of the piece again (The top bar), and the automation -- the moves, what the player's two hands do to the knobs over the whole piece. Every knob the composer moves has a lane: its offset from where you set the knob, as a curve over the piece, from -1 to +1 of the knob's range -- the filter of the bass row opening towards the peak and closing in the coda, the counter rows' filters, the lead's, the echo's feedback thrown now and then, the wind rising and falling in the atmosphere. The colour is the hand that moves the knob.

Above the lanes, a bar per hand shows where it is busy. A player has two hands, and the composer keeps to that: never more than two knobs move at once, and a hand finishes one move before it starts the next. The sections of the piece lie behind the lanes, the playhead runs over them, and a click jumps there. The moves can be held on the Perform page (Hold Moves): every knob then stays where the curve has it.

Voices



Each row plays its own modular voice: two drifting oscillators (a sawtooth blended into a pulse, the second detuned), a mixer driven into a soft saturation, a four-pole transistor ladder (or one of nine other filters, see Filters) at twice the sample rate, envelopes for the amplitude and the filter and a third for modulation, four LFOs and a modulation matrix (see Modulation), and glide. The oscillators run freely and are never reset on a note, and each wanders slowly in pitch, as analogue oscillators do; the Drift knob sets how far.

Instead of the two analogue oscillators a row can read a wavetable (Wavetable; Analog is the oscillators): the same tables as the pad synth's -- the PPG's, the sync and formant sweeps, the glass, vocal and sampled ones -- read band-limited by two oscillators with the row's detune and drift, into the same mixer and ladder. Table Position chooses the place in the table, and Table Mod how far the row's modulation sequencer moves it from note to note: a bright step reads further into the table, so the timbre wanders against the notes

as on Stürtzer's Iridium sequences. The composer gives the counter rows after the main sequence a table now and then (Modern in three rows of five, Doom in one), and in Modern sometimes the main sequence.

Presets. Every synth page -- the voices, the lead, the drone, the tape keys, the strings, the drums and the atmosphere -- has 1024 factory presets above its knobs, in sixteen groups of sixty-four ("Ladder Bass", "Glass Arp", "Cathedral Choir", "Slow Swell", "Solar Wind" ...); the arrows step through them. Each name is two words: the first runs from dark to bright ("Umbral" ... "Radiant"), the second is the group's own, so "Umbral Keel" is the darkest and "Radiant Piston" the brightest of the ladder basses. On the voices page a preset goes to the row chosen on the left. The disk keeps the synth's knobs as they stand under a name of your own (a small text file in the application data, Ephemeris/Presets); they appear under "User". All rows, on the voices page, sends a preset to every row at once. The effect pages (Echo + Spring, Hall) keep presets of your own too, each the whole page -- all its rooms -- as it stands. A preset sets the sound, never the mix: level, pan, the sends, the low cut, the distance and the drone's auto pan stay, and so do the atmosphere's amounts, which the composer moves. Nor does it put a synth out of tune with the rest: the drone and the bass groups keep their oscillators within a cent or three of each other (the pure foundation), the others within twelve, the lead's vibrato and the tapes' wobble stay inside their usual bounds, and no preset transposes anything.

Every piece has its own sounds. With Composer's Sounds on (the Master page; on by default) the composer chooses a factory preset for every synth in every piece, from the groups that fit the part: the bass row from Ladder Bass, Deep Ostinato, Dark Throb or Warm Unison, kept dark enough; the main sequence from Squelch Arp, Resonant Sweep, Pluck Sequence, Tape Sequence, Staccato Pulse or Accent Ratchet, with resonance and a deep envelope; the other counter rows from the glass, hollow and plucked groups; the lead and the kit by the style, the tape keys from the style's tape set -- darker in Doom, brighter in Melodic. The preset bar of each synth page shows "this piece: ..." with the name where the piece now plays, the mixer shows it under each strip's name, and reroll sounds draws others while every note stays. The composer sets the preset on the synth's knobs as the piece begins, as a program change: the page shows what plays, and a knob you turn moves from the preset's value. A synth whose sound a new piece keeps -- after a reroll of the rows or the form, say -- keeps your knobs as you left them; a new sound replaces them. In a concert each piece brings its sounds as it begins (a jump into another piece takes that piece's, a jump inside one leaves your knobs). A saved set and a saved session keep the knobs as they were, the sounds among them. Off, every piece plays your knobs as they are.

The knobs the player's hands move during a piece -- cutoff, resonance, envelope amount, decay -- are moved by the composer's gestures as offsets from where the piece's settings put them, so the ranges you choose keep their meaning.

VOICE (8 INSTANCES)

Control	Key	Range	Default
Wave	`voiceN.wave`	0 .. 1	0
Detune	`voiceN.detune`	0 .. 30 ct	7 ct
Pulse Width	`voiceN.pw`	0.05 .. 0.5	0.5
Drift	`voiceN.drift`	0 .. 15 ct	3 ct
Drive	`voiceN.drive`	0 .. 24 dB	6 dB
Cutoff	`voiceN.cutoff`	30 .. 16000 Hz	650 Hz
Resonance	`voiceN.resonance`	0 .. 1	0.4
Env Amount	`voiceN.env_amount`	0 .. 6 oct	3.5 oct
Filter Decay	`voiceN.decay`	15 .. 2000 ms	190 ms
Key Track	`voiceN.keytrack`	0 .. 1	0.5
Accent	`voiceN.accent`	0 .. 1	0.5
Amp Release	`voiceN.amp_decay`	5 .. 2000 ms	60 ms
Glide	`voiceN.glide`	1 .. 500 ms	60 ms
Wavetable	`voiceN.table`	Analog, Classic, PWM, Sync, Formant, Vocal, Organ, Glass, Metal, Bowed 1, Bowed 2, Bowed 3, Tube 1, Tube 2, Tube 3, Alto, Tenor, Soprano, Basso, Consonant 1, Consonant 2, Consonant 3, Overtones 1, Overtones 2, Overtones 3, Morph 1, Morph 2, Morph 3, Sampled 1, Sampled 2, Sampled 3, FM Synth, E-Organ, Violin, Theremin, PPG 00, PPG 03, PPG 06, PPG 07, PPG 12, PPG 17, PPG 22, PPG 25, PPG Upper, Drone Bank, Organic, Vox Synth, 921 Waves, Prophet Waves, SEM Waves, 2600 Waves, E-mu Waves	Analog
Table Position	`voiceN.table_pos`	0 .. 1	0
Table Mod	`voiceN.table_mod`	0 .. 1	0.4

Filter	`voiceN.filter`	Moog Ladder, Prophet (SSM2040/CEM3320), Juno (IR3109), Oberheim SEM, Xpander (pole mix), Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Moog Ladder
Filter Mode	`voiceN.filter_mode`	0 .. 1	0
Filter FM	`voiceN.filter_fm`	0 .. 1	0
Amp Attack	`voiceN.amp_attack`	0.5 .. 8000 ms	2 ms
Amp Decay	`voiceN.amp_decay2`	5 .. 8000 ms	50 ms
Amp Sustain	`voiceN.amp_sustain`	0 .. 1	1
Filter Attack	`voiceN.filt_attack`	0.5 .. 8000 ms	1.5 ms
Filter Sustain	`voiceN.filt_sustain`	0 .. 1	0
Filter Release	`voiceN.filt_release`	5 .. 8000 ms	180 ms
Release = Decay	`voiceN.filt_link`	off / on	on
Env Velocity	`voiceN.env_velocity`	0 .. 1	0
Mod Attack	`voiceN.mod_attack`	0.5 .. 8000 ms	10 ms
Mod Decay	`voiceN.mod_decay`	5 .. 8000 ms	500 ms
Mod Sustain	`voiceN.mod_sustain`	0 .. 1	0
Mod Release	`voiceN.mod_release`	5 .. 8000 ms	300 ms
LFO 1 Rate	`voiceN.lfo1_rate`	0.01 .. 30 Hz	0.5 Hz
LFO 1 Shape	`voiceN.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`voiceN.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`voiceN.lfo1_retrig`	off / on	off
LFO 1 Fade	`voiceN.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`voiceN.lfo2_rate`	0.01 .. 30 Hz	5 Hz
LFO 2 Shape	`voiceN.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`voiceN.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`voiceN.lfo2_retrig`	off / on	off
LFO 2 Fade	`voiceN.lfo2_fade`	0 .. 10 s	0 s
LFO 3 Rate	`voiceN.lfo3_rate`	0.01 .. 30 Hz	0.07 Hz
LFO 3 Shape	`voiceN.lfo3_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	`voiceN.lfo3_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 3 Retrig	`voiceN.lfo3_retrig`	off / on	off
LFO 3 Fade	`voiceN.lfo3_fade`	0 .. 10 s	0 s
LFO 4 Rate	`voiceN.lfo4_rate`	0.01 .. 30 Hz	0.2 Hz
LFO 4 Shape	`voiceN.lfo4_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	`voiceN.lfo4_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 4 Retrig	`voiceN.lfo4_retrig`	off / on	off
LFO 4 Fade	`voiceN.lfo4_fade`	0 .. 10 s	0 s

Mod 1 Source	`voiceN.mod1_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 1 Target	`voiceN.mod1_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 1 Amount	`voiceN.mod1_amt`	-1 .. 1	0
Mod 2 Source	`voiceN.mod2_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 2 Target	`voiceN.mod2_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 2 Amount	`voiceN.mod2_amt`	-1 .. 1	0
Mod 3 Source	`voiceN.mod3_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 3 Target	`voiceN.mod3_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 3 Amount	`voiceN.mod3_amt`	-1 .. 1	0
Mod 4 Source	`voiceN.mod4_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 4 Target	`voiceN.mod4_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 4 Amount	`voiceN.mod4_amt`	-1 .. 1	0
Mod 5 Source	`voiceN.mod5_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 5 Target	`voiceN.mod5_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 5 Amount	`voiceN.mod5_amt`	-1 .. 1	0
Mod 6 Source	`voiceN.mod6_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 6 Target	`voiceN.mod6_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 6 Amount	`voiceN.mod6_amt`	-1 .. 1	0
Mod 7 Source	`voiceN.mod7_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 7 Target	`voiceN.mod7_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 7 Amount	`voiceN.mod7_amt`	-1 .. 1	0
Mod 8 Source	`voiceN.mod8_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 8 Target	`voiceN.mod8_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 8 Amount	`voiceN.mod8_amt`	-1 .. 1	0
VCO	`voiceN.vco`	Analog, Moog 921, Prophet-5, Oberheim SEM, ARP 2600, E-mu Modular	Analog
Sync	`voiceN.sync`	off / on	off
Osc 2 Pitch	`voiceN.osc2_pitch`	-12 .. 24 st	0 st
Cross Mod	`voiceN.cross_mod`	0 .. 1	0

Filters

Every voice -- the rows, the lead, the drone -- and the pad synth choose their filter from ten models of the classic voltage-controlled filters (Filter on the synth's page, in the group Filter):

```
Moog Ladder         the transistor ladder, 24 dB: round, fat, the bass and the main sequence
Prophet             the OTA cascade of the SSM2040 and CEM3320 (and its reissue SSI2140, the Microwave's CEM3389):
                    present, its resonance saturating before its stages
Juno                the IR3109 of the Juno and Jupiter: cleaner, the pass band thinner with resonance
Oberheim SEM        the state-variable filter, 12 dB; Filter Mode morphs low pass -> notch -> high pass
Xpander             the pole mixing of the Xpander and Matrix-12 on the cascade; Filter Mode chooses the response:
                    LP4, LP2, BP2, BP4, HP2, HP4, notch, phaser
Diode Ladder        the TB-303's and the EMS's: squelchy, its resonance rubbery
Korg35              the MS-20's Sallen-Key with its diode-limited feedback: screaming at high resonance
Polivoks            the Soviet filter whose op-amps are slew-limited on purpose: gritty; Filter Mode to band pass
EDP Wasp            CMOS inverters as integrators: asymmetric, raw
Comb                a delay line tuned to the cutoff, its feedback damped; Filter Mode flips its sign
```

Filter FM lets the first oscillator move the cutoff at audio rate, as on the Xpander: metallic, bell-like edges.

How they are made. Each model is its circuit's differential equation, with the nonlinearities where the circuit has them -- the transistor pairs of a ladder, the OTAs of a cascade, the diode pairs of the diode ladder, the diode limiter in the Korg35's feedback, the slew-limited op-amps of the Polivoks, the inverters of the Wasp. The capacitors are integrated by the trapezoidal rule in the form of the topology-preserving transform (Zavalishin, "The Art of VA Filter Design"; Simper's equivalent capacitor currents), and the implicit equation this gives is solved every sample by Newton-Raphson on its exact Jacobian -- the delay-free loop resolved rather than broken by a sample's delay (D'Angelo and Välimäki 2014 for the ladder; Holters and Zölzer 2015 for nonlinear state-space models of circuits; Stinchcombe's analyses of the ladder, the diode ladder and the MS-20). The voices run at twice the sample rate, which keeps the aliasing of the saturating stages low. In the small-signal range each model follows its circuit's linear response to within 0.15 dB; at full resonance the ladders, the cascades, the diode ladder and the Korg35 oscillate at their cutoff.

The factory presets carry the filters of their instruments: the basses on the Moog ladder and now and then the diode ladder, the squelch on the diode ladder and the Korg35, glass on the Xpander's band and high passes, the Juno strings on the Juno, the Oberheim brass on the SEM, the dark and dusty sounds on the Polivoks and the Wasp, the cosmic drips on the comb and the phaser -- their cutoffs set so a group keeps its brightness whatever it drew.

Oscillators

Every voice -- the rows, the lead, the drone -- chooses its oscillators (VCO, in the group Oscillators): its own analog pair, or one of five classic VCOs.

```
Analog          the voices' own: ideal ramps and pulses
Moog 921        the Moog modular's oscillator: a discrete reset whose soft knee rounds the ramp's top; its pitch
                wanders the most and the slowest
Prophet-5       the SSM2030 of the Prophet-5: an integrated, nearly ideal core, steady -- and the sync sweeps
Oberheim SEM    a discrete core whose leakage bends the ramp towards the capacitor's charging curve: round, warm
ARP 2600        the 4027's fast reset, the ramp speeding up a little towards the top; the most jitter
E-mu Modular    precise and stable: ideal ramps, half the others' drift, a tenth of their jitter
```

Every one of them is a sawtooth core -- a capacitor charged by the exponential converter's current until a comparator fires and a transistor empties it -- and a comparator on the ramp for the pulse. What sets the models apart is what the circuit does to that: the ramp's bow (the core's leakage), the knee at its top (the reset transistor starting to conduct early), and above all how the pitch moves -- the drift's spread and speed, how far a note lands off its pitch, and a fast jitter of a fraction of a cent correlated over a few milliseconds. Honestly: the waveforms differ by a few tenths of a dB in the upper harmonics and in the phases of the lower ones; what the ear notices first is the wandering. The magnitudes follow the circuits' principles and the instruments' reputations; they are not measured on the instruments.

Sync restarts VCO 2 with every cycle of VCO 1 (hard sync, the Prophet's Osc A to Osc B). Osc 2 Pitch sets VCO 2's interval, -12 to +24 semitones; with sync it moves the timbre instead of the pitch, and the matrix's target Osc 2 Pitch from the Filter Env is the Prophet's Poly-Mod sweep -- the tearing, vocal sync lead. Cross Mod puts VCO 1 on VCO 2's frequency at audio rate, up to three octaves: the clangorous edges of the ARP 2600's FM and of the Poly-Mod from Osc B.

How they are made. The waveform's every discontinuity -- the reset, the pulse's edges, a sync restart -- is found exactly inside its sample step, and the jump and the change of slope there are smoothed with the polynomial BLEP and its integral, the BLAMP (Välimäki, Pekonen and Nam 2012; Esqueda, Välimäki and Bilbao 2016), at the voices' twice the sample rate. A synced tone of 1.2 kHz keeps its aliases 53 dB under its harmonics in the audible band (a naive restart: 19 dB). A voice on a model, with sync or cross mod runs its oscillators beside the SIMD lanes; the Analog pair stays in them.

The factory presets carry the oscillators of their instruments: the basses on the Moog (the deep ones now and then on the steady E-mu), the sync sweeps on the Prophet (Bright Stab, Screaming Filter, Cosmic Siren), the round pulses and the soft pads on the SEM, the plucks and the squelch on the ARP, glass and bells on the E-mu, some with cross mod. The pad synth's analog pads read the models' waves (see Poly).

Modulation

Every synth has full envelopes, LFOs and a modulation matrix: the rows' voices, the lead, the drone and the pad synth the whole set, the tape keys and the strings a smaller one.

Envelopes. Amp Envelope: Attack, Decay, Sustain and Release (on the voices the Release is the Amp Release that was there before). Filter Envelope: Attack, Decay -- the large knob, the one the hands move -- Sustain, Release, and Release = Decay: on, as by default, the release takes the decay's time, as with the Minimoog's decay switch, and with no sustain the decay simply runs on through the note's end, so a sequencer's short gates leave the filter's decay to the hand; off, the Release knob counts. Env Velocity lets the note's velocity scale the filter envelope. On the pad synth the filter follows the amplitude envelope (Filter Follows Amp) until you switch it off; then it has four times of its own. Mod Envelope is a third ADSR, for the matrix.

LFOs. Four per voice (two on the tape keys and the strings): Rate; Shape -- sine, triangle, saw up, saw down, square, sample & hold, smooth random; Sync -- free at Rate, or a division of the bar: 4, 2 or 1 bars, 1/2 to 1/16, quarter and eighth triplets, and a synced LFO keeps its phase on the piece's beat, so a one-bar sweep opens on every downbeat; Retrig -- every note starts it from the beginning; Fade -- it comes in over that many seconds after each note, as a hand reaching for the wheel. Every voice has its own LFOs, and on the pad synth every key has its own four, so the keys of a chord move apart.

The matrix. Eight slots (four on the tape keys and the strings), each a source, a target and an amount from -1 to 1 (double-click: none). The sources: LFO 1 to 4, Mod Env, Filter Env, Velocity, and Mod Lane -- the step of the row's modulation sequencer under the note. The targets, and what an amount of 1 means:

```
Pitch           12 semitones; the amount counts squared, so the first tenth of the knob holds a vibrato of a few cents
Pulse Width     0.45 of the duty cycle
Wave            the whole blend from saw to pulse
Table Position  the whole wavetable
Cutoff          five octaves
Resonance       the whole knob
Filter Mode     the SEM's morph, the Polivoks' band pass, the comb's sign (the Xpander's response stays the knob's)
Filter FM       the whole knob
Level           doubled at +1, shut at -1
Pan             from one side to the other
```

Slots on the same target add up. A target a synth does not have does nothing: the pad synth has no pulse width, blend or filter FM; the tape keys and the strings reach Pitch, Tone, Level and Pan. The tape keys' envelope lies over the machine's own: Swell (at its minimum, 1 ms, only the pressure pad's rise), Decay, Sustain Level, and Release -- the fall after the key, the machine's 70 ms by default; the tapes still run out after eight seconds. The strings keep their Crescendo and Release and have a Decay and a Sustain Level between them.

Everything runs on the piece's clock and the synth's own count of samples, never on the host's blocks: the envelopes every sample, the LFOs and the matrix at the synth's control steps (every 4 samples on the voices, 16 on the pad synth, 32 on the tape keys and the strings), and what they move reaches the voices per sample; the random shapes draw from a stream of their own, so modulation never shifts a voice's drift. A render is the same to the bit for any block size, and with the matrix empty every synth sounds exactly as it did without it.

The factory presets use it generously: nine in ten carry modulation, on average two slots of the matrix, over ten different targets on the voices. Each group has its own -- the modulation that makes it what it is -- and every preset draws one to three more from its synth's pool: the cutoff drifting or pulsing with the bars, sample and hold, the velocity and the row's modulation lane on filter and timbre, a pluck of the mod envelope, pulse width modulation, a wobble or a late vibrato, the pan wandering, the SEM's morph and the resonance breathing, a gate of the level in time, a zap of filter FM; on the pad synth the table walking and every key its own place and pitch. Every one gets the next free LFO and slot, so they never step on each other; the foundation -- the bass groups and the drone -- never moves its pitch. Among the groups' own: plucks and stabs decay under a held gate; the hollow pulses and the humming drones get their pulse width modulation; the resonant sweeps a filter sweep synced to two or four bars; the cosmic drips a sample-and-hold on the cutoff or a pitch blip from the mod envelope; the tape sequences and the dusty solos a wobble; the legato glides, the bowed pads and the violins a vibrato that comes in late; the PPG and morph pads a slow walk through their tables; the Oberheim brass a filter swell of its own. The composer's sounds bring them into a piece with the presets.

Lead



The lead is the same voice as a row's, with vibrato that comes in over the first half second of a held note, as a hand reaches the wheel. It plays in its own section and over the peak: phrases of four to eight bars with rests of two to four between them, on the pentatonic of the moment (with the sixth where the mode is Dorian), chord tones on the strong beats, glides from the tone below, repeats of its own phrases, and an end on the fifth or the minor third rather than the root. It plays a little behind the beat, as a player over a machine does.

LEAD

Control	Key	Range	Default

Wave	`lead.wave`	0 .. 1	0.25
Detune	`lead.detune`	0 .. 30 ct	4 ct
Pulse Width	`lead.pw`	0.05 .. 0.5	0.45
Drift	`lead.drift`	0 .. 15 ct	2 ct
Drive	`lead.drive`	0 .. 24 dB	8 dB
Cutoff	`lead.cutoff`	30 .. 16000 Hz	1500 Hz
Resonance	`lead.resonance`	0 .. 1	0.25
Env Amount	`lead.env_amount`	0 .. 6 oct	2 oct
Filter Decay	`lead.decay`	15 .. 2000 ms	450 ms
Key Track	`lead.keytrack`	0 .. 1	0.6
Accent	`lead.accent`	0 .. 1	0.3
Amp Release	`lead.amp_decay`	5 .. 2000 ms	300 ms
Glide	`lead.glide`	1 .. 500 ms	80 ms
Level	`lead.level`	-60 .. 6 dB	-9 dB
Pan	`lead.pan`	-1 .. 1	0.15
Echo Send	`lead.echo`	0 .. 1	0.45
Vibrato	`lead.vibrato`	0 .. 100 ct	18 ct
Vibrato Rate	`lead.vibrato_rate`	1 .. 10 Hz	5.2 Hz
Reverb Send	`lead.reverb`	0 .. 1	0.4
Auto Pan	`lead.auto_pan`	0 .. 1	0
Low Cut	`lead.low_cut`	10 .. 500 Hz	150 Hz
Distance	`lead.distance`	0 .. 1	0
Blend Send	`lead.blend`	0 .. 1	0
Early Send	`lead.early`	0 .. 1	0
Shimmer Send	`lead.shimmer`	0 .. 1	0
Filter	`lead.filter`	Moog Ladder, Prophet (SSM2040/CEM3320), Juno (IR3109), Oberheim SEM, Xpander (pole mix), Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Moog Ladder
Filter Mode	`lead.filter_mode`	0 .. 1	0
Filter FM	`lead.filter_fm`	0 .. 1	0
Amp Attack	`lead.amp_attack`	0.5 .. 8000 ms	2 ms
Amp Decay	`lead.amp_decay2`	5 .. 8000 ms	50 ms
Amp Sustain	`lead.amp_sustain`	0 .. 1	1
Filter Attack	`lead.filt_attack`	0.5 .. 8000 ms	1.5 ms
Filter Sustain	`lead.filt_sustain`	0 .. 1	0
Filter Release	`lead.filt_release`	5 .. 8000 ms	180 ms
Release = Decay	`lead.filt_link`	off / on	on
Env Velocity	`lead.env_velocity`	0 .. 1	0
Mod Attack	`lead.mod_attack`	0.5 .. 8000 ms	10 ms
Mod Decay	`lead.mod_decay`	5 .. 8000 ms	500 ms
Mod Sustain	`lead.mod_sustain`	0 .. 1	0
Mod Release	`lead.mod_release`	5 .. 8000 ms	300 ms
LFO 1 Rate	`lead.lfo1_rate`	0.01 .. 30 Hz	0.5 Hz
LFO 1 Shape	`lead.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`lead.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`lead.lfo1_retrig`	off / on	off
LFO 1 Fade	`lead.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`lead.lfo2_rate`	0.01 .. 30 Hz	5 Hz

LFO 2 Shape	`lead.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`lead.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`lead.lfo2_retrig`	off / on	off
LFO 2 Fade	`lead.lfo2_fade`	0 .. 10 s	0 s
LFO 3 Rate	`lead.lfo3_rate`	0.01 .. 30 Hz	0.07 Hz
LFO 3 Shape	`lead.lfo3_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	`lead.lfo3_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 3 Retrig	`lead.lfo3_retrig`	off / on	off
LFO 3 Fade	`lead.lfo3_fade`	0 .. 10 s	0 s
LFO 4 Rate	`lead.lfo4_rate`	0.01 .. 30 Hz	0.2 Hz
LFO 4 Shape	`lead.lfo4_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	`lead.lfo4_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 4 Retrig	`lead.lfo4_retrig`	off / on	off
LFO 4 Fade	`lead.lfo4_fade`	0 .. 10 s	0 s
Mod 1 Source	`lead.mod1_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 1 Target	`lead.mod1_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 1 Amount	`lead.mod1_amt`	-1 .. 1	0
Mod 2 Source	`lead.mod2_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 2 Target	`lead.mod2_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 2 Amount	`lead.mod2_amt`	-1 .. 1	0
Mod 3 Source	`lead.mod3_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 3 Target	`lead.mod3_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 3 Amount	`lead.mod3_amt`	-1 .. 1	0
Mod 4 Source	`lead.mod4_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 4 Target	`lead.mod4_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 4 Amount	`lead.mod4_amt`	-1 .. 1	0
Mod 5 Source	`lead.mod5_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 5 Target	`lead.mod5_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 5 Amount	`lead.mod5_amt`	-1 .. 1	0
Mod 6 Source	`lead.mod6_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 6 Target	`lead.mod6_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 6 Amount	`lead.mod6_amt`	-1 .. 1	0
Mod 7 Source	`lead.mod7_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 7 Target	`lead.mod7_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 7 Amount	`lead.mod7_amt`	-1 .. 1	0
Mod 8 Source	`lead.mod8_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 8 Target	`lead.mod8_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off

Mod 8 Amount	`lead.mod8_amt`	-1 .. 1	0
VCO	`lead.vco`	Analog, Moog 921, Prophet-5, Oberheim SEM, ARP 2600, E-mu Modular	Analog
Sync	`lead.sync`	off / on	off
Osc 2 Pitch	`lead.osc2_pitch`	-12 .. 24 st	0 st
Cross Mod	`lead.cross_mod`	0 .. 1	0

Drone



The drone is the ground under the whole piece: the root, low, held from the start to the fading coda, following the key when a phase changes it. It is a modular voice as well and has the lead's table. Auto Pan lets it wander slowly across the stereo field, on a sine of 0.05 Hz (twenty seconds from one side to the other and back); the lead has the knob as well, at zero. The drone's two oscillators are not detuned against each other, nor are the bass row's: the foundation of a piece stays in pure octaves and fifths, without a slow cloud of beating in the middle over the minutes; the detune belongs to the counter rows, the lead and the pads.

DRONE

Control	Key	Range	Default

Wave	`drone.wave`	0 .. 1	0
Detune	`drone.detune`	0 .. 30 ct	0 ct
Pulse Width	`drone.pw`	0.05 .. 0.5	0.5
Drift	`drone.drift`	0 .. 15 ct	4 ct
Drive	`drone.drive`	0 .. 24 dB	4 dB
Cutoff	`drone.cutoff`	30 .. 16000 Hz	320 Hz
Resonance	`drone.resonance`	0 .. 1	0.3
Env Amount	`drone.env_amount`	0 .. 6 oct	0.5 oct
Filter Decay	`drone.decay`	15 .. 2000 ms	900 ms
Key Track	`drone.keytrack`	0 .. 1	0.3
Accent	`drone.accent`	0 .. 1	0
Amp Release	`drone.amp_decay`	5 .. 2000 ms	2000 ms
Glide	`drone.glide`	1 .. 500 ms	300 ms
Level	`drone.level`	-60 .. 6 dB	-10 dB
Pan	`drone.pan`	-1 .. 1	0
Echo Send	`drone.echo`	0 .. 1	0.1
Vibrato	`drone.vibrato`	0 .. 100 ct	0 ct
Vibrato Rate	`drone.vibrato_rate`	1 .. 10 Hz	5 Hz
Reverb Send	`drone.reverb`	0 .. 1	0.55
Auto Pan	`drone.auto_pan`	0 .. 1	0.35
Low Cut	`drone.low_cut`	10 .. 500 Hz	40 Hz
Distance	`drone.distance`	0 .. 1	0
Blend Send	`drone.blend`	0 .. 1	0
Early Send	`drone.early`	0 .. 1	0
Shimmer Send	`drone.shimmer`	0 .. 1	0
Filter	`drone.filter`	Moog Ladder, Prophet (SSM2040/CEM3320), Juno (IR3109), Oberheim SEM, Xpander (pole mix), Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Moog Ladder
Filter Mode	`drone.filter_mode`	0 .. 1	0
Filter FM	`drone.filter_fm`	0 .. 1	0
Amp Attack	`drone.amp_attack`	0.5 .. 8000 ms	2 ms
Amp Decay	`drone.amp_decay2`	5 .. 8000 ms	50 ms
Amp Sustain	`drone.amp_sustain`	0 .. 1	1
Filter Attack	`drone.filt_attack`	0.5 .. 8000 ms	1.5 ms
Filter Sustain	`drone.filt_sustain`	0 .. 1	0
Filter Release	`drone.filt_release`	5 .. 8000 ms	180 ms
Release = Decay	`drone.filt_link`	off / on	on
Env Velocity	`drone.env_velocity`	0 .. 1	0
Mod Attack	`drone.mod_attack`	0.5 .. 8000 ms	10 ms
Mod Decay	`drone.mod_decay`	5 .. 8000 ms	500 ms
Mod Sustain	`drone.mod_sustain`	0 .. 1	0
Mod Release	`drone.mod_release`	5 .. 8000 ms	300 ms
LFO 1 Rate	`drone.lfo1_rate`	0.01 .. 30 Hz	0.5 Hz
LFO 1 Shape	`drone.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`drone.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`drone.lfo1_retrig`	off / on	off
LFO 1 Fade	`drone.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`drone.lfo2_rate`	0.01 .. 30 Hz	5 Hz

LFO 2 Shape	`drone.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`drone.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`drone.lfo2_retrig`	off / on	off
LFO 2 Fade	`drone.lfo2_fade`	0 .. 10 s	0 s
LFO 3 Rate	`drone.lfo3_rate`	0.01 .. 30 Hz	0.07 Hz
LFO 3 Shape	`drone.lfo3_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	`drone.lfo3_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 3 Retrig	`drone.lfo3_retrig`	off / on	off
LFO 3 Fade	`drone.lfo3_fade`	0 .. 10 s	0 s
LFO 4 Rate	`drone.lfo4_rate`	0.01 .. 30 Hz	0.2 Hz
LFO 4 Shape	`drone.lfo4_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	`drone.lfo4_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 4 Retrig	`drone.lfo4_retrig`	off / on	off
LFO 4 Fade	`drone.lfo4_fade`	0 .. 10 s	0 s
Mod 1 Source	`drone.mod1_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 1 Target	`drone.mod1_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 1 Amount	`drone.mod1_amt`	-1 .. 1	0
Mod 2 Source	`drone.mod2_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 2 Target	`drone.mod2_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 2 Amount	`drone.mod2_amt`	-1 .. 1	0
Mod 3 Source	`drone.mod3_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 3 Target	`drone.mod3_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 3 Amount	`drone.mod3_amt`	-1 .. 1	0
Mod 4 Source	`drone.mod4_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 4 Target	`drone.mod4_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 4 Amount	`drone.mod4_amt`	-1 .. 1	0
Mod 5 Source	`drone.mod5_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 5 Target	`drone.mod5_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 5 Amount	`drone.mod5_amt`	-1 .. 1	0
Mod 6 Source	`drone.mod6_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 6 Target	`drone.mod6_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 6 Amount	`drone.mod6_amt`	-1 .. 1	0
Mod 7 Source	`drone.mod7_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 7 Target	`drone.mod7_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off
Mod 7 Amount	`drone.mod7_amt`	-1 .. 1	0
Mod 8 Source	`drone.mod8_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 8 Target	`drone.mod8_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan, Osc 2 Pitch	Off

Mod 8 Amount	`drone.mod8_amt`	-1 .. 1	0
VCO	`drone.vco`	Analog, Moog 921, Prophet-5, Oberheim SEM, ARP 2600, E-mu Modular	Analog
Sync	`drone.sync`	off / on	off
Osc 2 Pitch	`drone.osc2_pitch`	-12 .. 24 st	0 st
Cross Mod	`drone.cross_mod`	0 .. 1	0

Tape Keys



A tape keyboard, synthesised: every key has its own strip of tape, with its own small detune, level and onset lag, played by one capstan whose speed wanders (wow and flutter) and sags a little under many keys. A tape runs out after eight seconds, as the instrument's did. Three sets: a choir of several singers through the formants of a sung vowel, strings, and flute. Age puts more of the machine's character into the sound; the tape set can change between pieces.

Spread widens the mono keyboard without colouring it: the side is the keyboard through a chain of all-passes whose phase turns at corners spread over the band, added on one side and taken away on the other, so the two sides differ in phase, not in tone, and their sum -- what a mono speaker plays -- is the keyboard alone. At the default 0.7 the correlation of the two sides is about 0.35: wide, with a middle the chords stand on.

TAPE

Control	Key	Range	Default
Tapes	`tape.set`	Choir, Strings, Flute	Choir
Vowel	`tape.vowel`	0 .. 1	0.2
Wow	`tape.wow`	0 .. 30 ct	6 ct
Flutter	`tape.flutter`	0 .. 10 ct	2 ct
Motor Load	`tape.sag`	0 .. 5 ct	1 ct
Tone	`tape.tone`	1500 .. 16000 Hz	7000 Hz
Age	`tape.age`	0 .. 1	0.5
Level	`tape.level`	-60 .. 6 dB	-1 dB
Pan	`tape.pan`	-1 .. 1	-0.1
Echo Send	`tape.echo`	0 .. 1	0.1
Reverb Send	`tape.reverb`	0 .. 1	0.5
Low Cut	`tape.low_cut`	10 .. 500 Hz	150 Hz
Distance	`tape.distance`	0 .. 1	0
Spread	`tape.spread`	0 .. 1	0.7
Blend Send	`tape.blend`	0 .. 1	0
Early Send	`tape.early`	0 .. 1	0
Shimmer Send	`tape.shimmer`	0 .. 1	0
Swell	`tape.amp_attack`	1 .. 8000 ms	1 ms
Decay	`tape.amp_decay`	5 .. 8000 ms	1500 ms
Sustain Level	`tape.amp_sustain`	0 .. 1	1
Release	`tape.amp_release`	10 .. 3000 ms	70 ms
LFO 1 Rate	`tape.lfo1_rate`	0.01 .. 30 Hz	5 Hz
LFO 1 Shape	`tape.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`tape.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`tape.lfo1_retrig`	off / on	off
LFO 1 Fade	`tape.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`tape.lfo2_rate`	0.01 .. 30 Hz	0.3 Hz
LFO 2 Shape	`tape.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`tape.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`tape.lfo2_retrig`	off / on	off
LFO 2 Fade	`tape.lfo2_fade`	0 .. 10 s	0 s
Mod 1 Source	`tape.mod1_src`	Off, LFO 1, LFO 2	Off
Mod 1 Target	`tape.mod1_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 1 Amount	`tape.mod1_amt`	-1 .. 1	0
Mod 2 Source	`tape.mod2_src`	Off, LFO 1, LFO 2	Off
Mod 2 Target	`tape.mod2_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 2 Amount	`tape.mod2_amt`	-1 .. 1	0
Mod 3 Source	`tape.mod3_src`	Off, LFO 1, LFO 2	Off
Mod 3 Target	`tape.mod3_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 3 Amount	`tape.mod3_amt`	-1 .. 1	0
Mod 4 Source	`tape.mod4_src`	Off, LFO 1, LFO 2	Off
Mod 4 Target	`tape.mod4_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 4 Amount	`tape.mod4_amt`	-1 .. 1	0

Strings



A string machine of the seventies, after the example of Waldorf's Streichfett: top-octave generators divided down into five registers per key -- sawtooths at 16', 8' and 4', squares at 8' and 4' -- through an ensemble of bucket-brigade delays swept by a slow and a fast modulation. Keys glide from where they were when a chord is stolen, as the machine's did.

Registration morphs continuously through eight mixes of the registers: Violins, Violas, Cellos, Basses, Full, Hollow, Brass and Organ, each with its own brightness and all equally loud. Animate lets a slow sine move the registration by itself, up to three and a half mixes either way at the Animate Rate, so the blend of the voices wanders under a held chord; the composer's hands move the registration now and then as well. Octave Balance tilts the footages towards the low or the high ones. Ensemble Type chooses the chorus: Solina

(three lines, a slow and a fast sweep), Chorus (two lines, one sweep) or Wide (three longer, deeper lines, spread wider). Phaser sends the machine through four swept all-passes, the classic swirl of strings through a phase shifter.

STRINGS

Control	Key	Range	Default
Crescendo	`strings.attack`	0.005 .. 3 s	0.35 s
Release	`strings.release`	0.05 .. 6 s	1.2 s
Octave Balance	`strings.feet`	0 .. 1	0.4
Tone	`strings.tone`	800 .. 12000 Hz	4500 Hz
Ensemble	`strings.ensemble`	0 .. 1	0.8
Level	`strings.level`	-60 .. 6 dB	-10 dB
Pan	`strings.pan`	-1 .. 1	0.1
Echo Send	`strings.echo`	0 .. 1	0.05
Reverb Send	`strings.reverb`	0 .. 1	0.45
Registration	`strings.registration`	0 .. 7	0
Animate	`strings.animate`	0 .. 1	0.3
Animate Rate	`strings.animate_rate`	0.01 .. 1 Hz	0.05 Hz
Ensemble Type	`strings.ensemble_type`	Solina, Chorus, Wide	Solina
Phaser	`strings.phaser`	0 .. 1	0
Low Cut	`strings.low_cut`	10 .. 500 Hz	200 Hz
Distance	`strings.distance`	0 .. 1	0
Blend Send	`strings.blend`	0 .. 1	0
Early Send	`strings.early`	0 .. 1	0
Shimmer Send	`strings.shimmer`	0 .. 1	0
Decay	`strings.amp_decay`	0.05 .. 20 s	2 s
Sustain Level	`strings.amp_sustain`	0 .. 1	1
LFO 1 Rate	`strings.lfo1_rate`	0.01 .. 30 Hz	5.5 Hz
LFO 1 Shape	`strings.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`strings.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`strings.lfo1_retrig`	off / on	off
LFO 1 Fade	`strings.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`strings.lfo2_rate`	0.01 .. 30 Hz	0.15 Hz
LFO 2 Shape	`strings.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`strings.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`strings.lfo2_retrig`	off / on	off
LFO 2 Fade	`strings.lfo2_fade`	0 .. 10 s	0 s
Mod 1 Source	`strings.mod1_src`	Off, LFO 1, LFO 2	Off
Mod 1 Target	`strings.mod1_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 1 Amount	`strings.mod1_amt`	-1 .. 1	0
Mod 2 Source	`strings.mod2_src`	Off, LFO 1, LFO 2	Off
Mod 2 Target	`strings.mod2_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 2 Amount	`strings.mod2_amt`	-1 .. 1	0
Mod 3 Source	`strings.mod3_src`	Off, LFO 1, LFO 2	Off
Mod 3 Target	`strings.mod3_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 3 Amount	`strings.mod3_amt`	-1 .. 1	0
Mod 4 Source	`strings.mod4_src`	Off, LFO 1, LFO 2	Off
Mod 4 Target	`strings.mod4_dst`	Off, Pitch, Tone, Level, Pan	Off
Mod 4 Amount	`strings.mod4_amt`	-1 .. 1	0

Poly



The pad synth: the polysynth plane of the modern Berlin School -- the Oberheim and Juno pads under Stürtzer's, Quaeschning's and Redshift's sequences, and the glassy planes of the PPG Wave and the Waldorf wavetable synths. Eight keys, each with two oscillators a few cents apart and spread across the stereo field, reading one wavetable; a 12 dB state-variable low pass (the Oberheim SEM's kind) that follows the key by half and the envelope by Env Amount; a slow envelope (the guide's attack of one to four seconds, release of three to eight); and an ensemble after it, one chorus line read on either side with its sweep turned against itself -- width that keeps its middle.

A wavetable is a stack of single cycles; Position chooses where in the stack the oscillators read, and Scan moves the position with a slow sine at Scan Rate, each key a little apart in its phase -- the one slow modulation a pad needs. On the PWM table (the default) the

position is the pulse width, so the scan is the pulse width modulation of an analog pad; on a PPG or Waldorf table the timbre walks through its frames. The tables are read band-limited at eight resolutions, so a high note never aliases. Fifty-one tables: eight from formulas -- Classic (sine, triangle, saw, square, pulse), PWM, Sync (a saw hard-synced to one to five times its master), Formant (a resonant peak climbing through a saw), Vocal, Organ, Glass and Metal -- and thirty-eight sampled single cycles from the AmbientSynth's library: its own bowed, tube, vocal, consonant, overtone, morph and sampled tables, nine of the PPG Wave's tables as the users of WaveEdit Online rebuilt them, and a few of the Adventure Kid waveforms (AKWF, by Kristoffer Ekstrand; WaveEdit Online and AKWF are CC0); and the waves of the five classic VCOs (921, Prophet, SEM, 2600 and E-mu Waves; see Oscillators), each model's ramp morphing into its square and then into a narrowing pulse -- the analog pads read them now and then instead of the PWM table.

The composer brings the pads in where the other planes leave room: in the atmosphere under the drone, from the first build until the tape keys come in, in the breakdowns and the bridges -- never as a third plane beside the tape keys and the strings, and an octave under the strings where they play (the production guide's rule). Four pieces in five of Drift have them, fewer than half of Doom. Its 1024 presets come in sixteen groups -- Analog Pad, Juno Strings, Oberheim Brass, Sync Sweep, Formant Pad, Vocal Pad, Glass Pad, Single Cycle Pad, Bowed Pad, Tube Pad, PPG Choir, PPG Upper, Overtone Pad, Morph Pad, Sampled Air, Dark Drone Pad -- and the composer's sounds choose among them by style.

POLY

Control	Key	Range	Default
Wavetable	`poly.table`	Classic, PWM, Sync, Formant, Vocal, Organ, Glass, Metal, Bowed 1, Bowed 2, Bowed 3, Tube 1, Tube 2, Tube 3, Alto, Tenor, Soprano, Basso, Consonant 1, Consonant 2, Consonant 3, Overtones 1, Overtones 2, Overtones 3, Morph 1, Morph 2, Morph 3, Sampled 1, Sampled 2, Sampled 3, FM Synth, E-Organ, Violin, Theremin, PPG 00, PPG 03, PPG 06, PPG 07, PPG 12, PPG 17, PPG 22, PPG 25, PPG Upper, Drone Bank, Organic, Vox Synth, 921 Waves, Prophet Waves, SEM Waves, 2600 Waves, E-mu Waves	PWM
Position	`poly.position`	0 .. 1	0.3
Scan	`poly.scan`	0 .. 1	0.35
Scan Rate	`poly.scan_rate`	0.005 .. 1 Hz	0.06 Hz
Detune	`poly.detune`	0 .. 30 ct	8 ct
Spread	`poly.spread`	0 .. 1	0.6
Drift	`poly.drift`	0 .. 12 ct	3 ct
Cutoff	`poly.cutoff`	100 .. 12000 Hz	2200 Hz
Resonance	`poly.resonance`	0 .. 1	0.15
Env Amount	`poly.env_amount`	0 .. 4 oct	1 oct
Attack	`poly.attack`	0.01 .. 8 s	1.5 s
Release	`poly.release`	0.1 .. 10 s	4 s
Ensemble	`poly.chorus`	0 .. 1	0.5
Level	`poly.level`	-60 .. 6 dB	-4 dB
Pan	`poly.pan`	-1 .. 1	-0.1
Echo Send	`poly.echo`	0 .. 1	0
Reverb Send	`poly.reverb`	0 .. 1	0.5
Low Cut	`poly.low_cut`	10 .. 500 Hz	180 Hz
Distance	`poly.distance`	0 .. 1	0
Blend Send	`poly.blend`	0 .. 1	0
Early Send	`poly.early`	0 .. 1	0
Shimmer Send	`poly.shimmer`	0 .. 1	0.1
Filter	`poly.filter`	Moog Ladder, Prophet (SSM2040/CEM3320), Juno (IR3109), Oberheim SEM, Xpander (pole mix), Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Oberheim SEM
Filter Mode	`poly.filter_mode`	0 .. 1	0
Amp Decay	`poly.amp_decay`	0.01 .. 20 s	1 s
Amp Sustain	`poly.amp_sustain`	0 .. 1	1

Filter Follows Amp	`poly.filt_link`	off / on	on
Filter Attack	`poly.filt_attack`	0.01 .. 20 s	2 s
Filter Decay	`poly.filt_decay`	0.01 .. 20 s	3 s
Filter Sustain	`poly.filt_sustain`	0 .. 1	0.4
Filter Release	`poly.filt_release`	0.01 .. 20 s	4 s
Env Velocity	`poly.env_velocity`	0 .. 1	0
Mod Attack	`poly.mod_attack`	0.01 .. 20 s	1 s
Mod Decay	`poly.mod_decay`	0.01 .. 20 s	2 s
Mod Sustain	`poly.mod_sustain`	0 .. 1	0.5
Mod Release	`poly.mod_release`	0.01 .. 20 s	3 s
LFO 1 Rate	`poly.lfo1_rate`	0.01 .. 30 Hz	0.2 Hz
LFO 1 Shape	`poly.lfo1_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	`poly.lfo1_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 1 Retrig	`poly.lfo1_retrig`	off / on	off
LFO 1 Fade	`poly.lfo1_fade`	0 .. 10 s	0 s
LFO 2 Rate	`poly.lfo2_rate`	0.01 .. 30 Hz	4.5 Hz
LFO 2 Shape	`poly.lfo2_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	`poly.lfo2_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 2 Retrig	`poly.lfo2_retrig`	off / on	off
LFO 2 Fade	`poly.lfo2_fade`	0 .. 10 s	0 s
LFO 3 Rate	`poly.lfo3_rate`	0.01 .. 30 Hz	0.05 Hz
LFO 3 Shape	`poly.lfo3_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	`poly.lfo3_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 3 Retrig	`poly.lfo3_retrig`	off / on	off
LFO 3 Fade	`poly.lfo3_fade`	0 .. 10 s	0 s
LFO 4 Rate	`poly.lfo4_rate`	0.01 .. 30 Hz	0.11 Hz
LFO 4 Shape	`poly.lfo4_shape`	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	`poly.lfo4_sync`	Free, 4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/4T, 1/8T	Free
LFO 4 Retrig	`poly.lfo4_retrig`	off / on	off
LFO 4 Fade	`poly.lfo4_fade`	0 .. 10 s	0 s
Mod 1 Source	`poly.mod1_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 1 Target	`poly.mod1_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 1 Amount	`poly.mod1_amt`	-1 .. 1	0
Mod 2 Source	`poly.mod2_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off

Mod 2 Target	`poly.mod2_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 2 Amount	`poly.mod2_amt`	-1 .. 1	0
Mod 3 Source	`poly.mod3_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 3 Target	`poly.mod3_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 3 Amount	`poly.mod3_amt`	-1 .. 1	0
Mod 4 Source	`poly.mod4_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 4 Target	`poly.mod4_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 4 Amount	`poly.mod4_amt`	-1 .. 1	0
Mod 5 Source	`poly.mod5_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 5 Target	`poly.mod5_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 5 Amount	`poly.mod5_amt`	-1 .. 1	0
Mod 6 Source	`poly.mod6_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 6 Target	`poly.mod6_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 6 Amount	`poly.mod6_amt`	-1 .. 1	0
Mod 7 Source	`poly.mod7_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 7 Target	`poly.mod7_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 7 Amount	`poly.mod7_amt`	-1 .. 1	0
Mod 8 Source	`poly.mod8_src`	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Mod Lane	Off
Mod 8 Target	`poly.mod8_dst`	Off, Pitch, Pulse Width, Wave, Table Position, Cutoff, Resonance, Filter Mode, Filter FM, Level, Pan	Off
Mod 8 Amount	`poly.mod8_amt`	-1 .. 1	0

Atmosphere



Wind (filtered noise in a moving band), slow sweeps, and bursts of sample-and-hold bleeps on the notes of the scale: the space at the start of a piece and in the bridges, and the stars over the builds.

Grains is a granular cloud: short tones, each a sine under a soft window of 40 to 250 ms, on the notes of the scale over the current root two to three octaves up, each at its own place in the stereo field, as many per second as Grain Density says. A piece has the cloud with the style's chance (the Style page); the hands then raise it in the atmosphere, the bridges and the coda, and take it away when the rows come in.

ATMOS

Control	Key	Range	Default
Wind	`atmos.wind`	-60 .. 0 dB	-60 dB
Wind Tone	`atmos.wind_tone`	150 .. 4000 Hz	700 Hz
Sweeps	`atmos.sweeps`	0 .. 6 /min	0 /min
Sweep Level	`atmos.sweep_level`	-60 .. 0 dB	-18 dB
Bleeps	`atmos.bleeps`	0 .. 12 /min	0 /min
Bleep Level	`atmos.bleep_level`	-60 .. 0 dB	-22 dB
Level	`atmos.level`	-60 .. 6 dB	0 dB
Echo Send	`atmos.echo`	0 .. 1	0.2
Reverb Send	`atmos.reverb`	0 .. 1	0.6
Grains	`atmos.grains`	-60 .. 0 dB	-60 dB
Grain Density	`atmos.grain_density`	1 .. 40 /s	12 /s
Low Cut	`atmos.low_cut`	10 .. 500 Hz	120 Hz
Shimmer Send	`atmos.shimmer`	0 .. 1	0

Echo + Spring



A tape echo with a moving read head -- wow, flutter, saturation and loss of highs in the feedback -- synchronised to the tempo, with a ping-pong option; its feedback can run into self-oscillation, which the echo throw uses. Springs are built into it, as in the tape echoes of the time: they take the echo's send and return beside it. The composer keeps the bass and the pads out of the echo -- the bass dry, the tape keys and the strings in the hall only -- and the hands lean the feedback high in the quiet parts of a piece and low at the peak. Low Cut is a high pass inside the loop, so every repeat is thinner than the one before, as a distant sound is; Duck lowers the return (of both echoes) while the rows play. The repeats of both echoes sit at about 70 % of the stereo base rather than hard left and right, so the middle keeps a share of each and the ping-pong survives in mono.

Echo switches the same send to a bucket-brigade delay instead of the tape: an analogue delay of 4096 stages as in the pedals of the late seventies. Its clock runs slower the longer the delay is set, and its filters have to follow the clock, so the longer the echo, the darker its repeats; the clock wanders slowly (Wow sets how far), and the compander saturates and hisses faintly in the loop. Time, feedback, tone, drive and ping-pong work as on the tape.

Echo 2 is a second, cleaner tape echo on a send of its own (the rows' Echo 2 Send), with its own time, feedback, tone, ping-pong and return; its repeats go into the hall as the first echo's do. It lets each layer of the sequence have its own delay: the main sequence on the first echo's dotted eighths, the other counter rows on the second in eighths or quarter triplets (1/4T, which both echoes offer), as the composer draws it for each piece.

ECHO

Control	Key	Range	Default
Time	`echo.time`	1/16, 1/8, 3/16, 1/4, 3/8, 1/2, 1/4T	3/16
Feedback	`echo.feedback`	0 .. 1.1	0.45
Tone	`echo.tone`	800 .. 12000 Hz	3500 Hz
Wow	`echo.wow`	0 .. 4 ms	0.2 ms
Flutter	`echo.flutter`	0 .. 0.5 ms	0.02 ms
Tape Drive	`echo.drive`	0 .. 18 dB	4 dB

Ping-Pong	`echo.pingpong`	off / on	on
Return	`echo.return`	-60 .. 6 dB	-4 dB
Echo	`echo.type`	Tape, BBD	Tape
Low Cut	`echo.low_cut`	20 .. 1000 Hz	200 Hz
Duck	`echo.duck`	0 .. 12 dB	3 dB

SPRING

Control	Key	Range	Default
Decay	`spring.decay`	0.3 .. 8 s	2.2 s
Tone	`spring.tone`	1500 .. 9000 Hz	4500 Hz
Return	`spring.return`	-60 .. 12 dB	-6 dB

DELAY

Control	Key	Range	Default
Echo 2 Time	`delay.time`	1/16, 1/8, 3/16, 1/4, 3/8, 1/2, 1/4T	1/8
Echo 2 Feedback	`delay.feedback`	0 .. 1	0.3
Echo 2 Tone	`delay.tone`	800 .. 12000 Hz	5000 Hz
Echo 2 Ping-Pong	`delay.pingpong`	off / on	on
Echo 2 Return	`delay.return`	-60 .. 6 dB	-6 dB
Echo 2 Low Cut	`delay.low_cut`	20 .. 1000 Hz	300 Hz

Hall



A long, dark hall (an eight-line feedback delay network) with its size, decay, damping, pre-delay and the band it returns -- by default 250 Hz to 6.5 kHz, so the hall adds space without mud or glare. The composer lets its decay grow in the atmosphere, the bridges and the coda and shrink by 40 % at the peak, and sets the pre-delay to a 64th note at the piece's tempo (32 ms at 118 BPM), which keeps the dry sounds in front of their room. Duck lowers the hall's return by up to that many dB while the rows play, so the tail blooms in the gaps and steps back under the notes.

The blend room is a second, short plate (Blend Decay about a second, a few ms of pre-delay, 240 Hz to 7 kHz) on a send of its own (Blend Send on the rows, the lead, the drone, the tape keys, the strings and the drums). Into Hall sends a share of its return on into the hall, so the far room sounds like the near one going on, not like a second place -- the serial far space of the dark-ambient practice. The composer puts the second layer there -- the counter rows, the lead, the drums -- with only a little hall; the pads, the drone and the atmosphere stay in the hall, the bass stays dry; at the peak the feed into the hall drops to 5 %, where the far room would fill up. Both returns are ducked by the rows alike.

With the early reflections (send A) and the shimmer (send D) the four rooms of the production guide are complete. The early reflections are a room without its tail: twelve reflections a side within Early Size (35 ms), 200 Hz to 8 kHz -- distance and glue for the counter rows, the lead and the drums, with nothing that smears. The shimmer is the effect hall: a long plate (Shimmer Decay, 8 s) whose output, shifted up an exact octave, is fed back into it (Shimmer, 0.35), so every round of the tail is an octave brighter and fainter; 400 Hz to 4 kHz. The composer opens it for the drone, the atmosphere and the tape keys in the atmosphere, the bridges and the coda, and closes it while the machine runs. Being an octave, it never detunes anything.

Room switches the same send to a plate instead: Dattorro's plate reverberator (1997) -- four diffusers and a tank of two halves that feed each other, each with a slowly moving all-pass, a damping low pass and the decay -- denser and brighter than the hall and without a room's early reflections, the sound of the plates on the records of the seventies. Decay, damping, pre-delay and the band work on it as well; its return is matched to the hall's level, so the switch changes the room, not the loudness.

REVERB

Control	Key	Range	Default
Size	`reverb.size`	0.3 .. 3	1.8
Decay	`reverb.decay`	0.3 .. 20 s	5.5 s
Damping	`reverb.damping`	0 .. 1	0.45
Pre-Delay	`reverb.predelay`	0 .. 200 ms	25 ms
Low Cut	`reverb.lowcut`	40 .. 500 Hz	250 Hz
High Cut	`reverb.highcut`	1000 .. 20000 Hz	6500 Hz
Return	`reverb.return`	-60 .. 12 dB	4 dB
Room	`reverb.type`	Hall, Plate	Hall
Duck	`reverb.duck`	0 .. 12 dB	2 dB

BLEND

Control	Key	Range	Default
Blend Decay	`blend.decay`	0.3 .. 4 s	1.2 s
Blend Pre-Delay	`blend.predelay`	0 .. 60 ms	8 ms
Blend Low Cut	`blend.low_cut`	60 .. 800 Hz	240 Hz
Blend High Cut	`blend.high_cut`	1500 .. 16000 Hz	7000 Hz
Blend Damping	`blend.damping`	0 .. 1	0.4
Blend Return	`blend.return`	-60 .. 6 dB	-2 dB
Into Hall	`blend.into_hall`	0 .. 0.5	0.15

EARLY

Control	Key	Range	Default
Early Size	`early.size`	10 .. 80 ms	35 ms
Early Low Cut	`early.low_cut`	60 .. 800 Hz	200 Hz
Early High Cut	`early.high_cut`	2000 .. 16000 Hz	8000 Hz
Early Return	`early.return`	-60 .. 6 dB	-4 dB

SHIMMER

Control	Key	Range	Default
Shimmer Decay	`shimmer.decay`	2 .. 20 s	8 s
Shimmer	`shimmer.amount`	0 .. 0.6	0.35
Shimmer Low Cut	`shimmer.low_cut`	100 .. 1500 Hz	400 Hz
Shimmer High Cut	`shimmer.high_cut`	1500 .. 12000 Hz	4000 Hz
Shimmer Return	`shimmer.return`	-60 .. 6 dB	-8 dB

Drums



A small analogue kit for the styles that have drums: kick, snare, hats, toms tuned to the root, rim and shaker, with patterns per style. The kit comes late in a piece, from the second build to the end of the peak, and never before 30 % of the piece or after 90 %: the sequence is there before it and stays after it.

Before the kit, a pulse carries the sequence in most pieces: a soft kick four to the floor in Melodic and Modern, a heartbeat in Cosmic, a slow beat on one and three in Drift, a low tom once a bar in Doom. It comes with the second build (not before a fifth of the piece) and makes way for the kit or the breakdown; Melodic has it in four pieces out of five, Drift in one out of three.

DRUMS

Control	Key	Range	Default
Kick Pitch	`drums.kick_hz`	35 .. 80 Hz	50 Hz
Decay	`drums.decay`	0.3 .. 2	1
Tone	`drums.tone`	0 .. 1	0.5
Level	`drums.level`	-60 .. 6 dB	-9 dB
Echo Send	`drums.echo`	0 .. 1	0.05
Reverb Send	`drums.reverb`	0 .. 1	0.2
Low Cut	`drums.low_cut`	10 .. 500 Hz	25 Hz
Blend Send	`drums.blend`	0 .. 1	0
Early Send	`drums.early`	0 .. 1	0

Master



The master: its level, a gentle bus compressor that keeps the music's dynamics (a ratio of at most 1.5), and a true-peak limiter as protection at -1 dBTP. Before them the mix bus removes DC and subsonics under 20 Hz, keeps everything under 100 Hz in mono whatever its pan, and widens what lies above 300 Hz by Width (dB of side) -- never so far that the side comes within 3 dB of the mid, so the correlation stays positive and the mix survives in mono. Mono sums both sides after the limiter, to check the mix as a small speaker would play it. Soft Clip rounds the peaks above the ceiling into at most that many dB before the limiter (with anti-aliasing), so short ratchets do not make the limiter work; quieter material passes untouched. Sub Ceiling holds the band under 80 Hz on its own under that level (a Linkwitz-Riley split), so the sub never pushes the whole mix into the limiter. The composer moves the level along the form instead of compressing it -- the spaces 3 dB under the peak, the entry and the breakdown 1.5 dB -- and the width too: widest in the spaces, narrow where the dry bass comes in, growing again with the layers; while the lead plays, the tape keys and the strings step back 3 dB.

The styles are levelled to the loudness a streamed release is played at (EBU R128, measured with the offline renderer on ten-minute pieces): Cosmic, Melodic and Modern about -15 LUFS integrated, Doom and Drift about -17.5. Since the pieces are drawn anew each time -- how many rows, which presets, whether the drums come in -- two pieces of one style could lie four dB apart; now every piece's loudest part is measured and brought to its style's level (at most 4 dB either way): the composer marks the peak, the program renders twenty seconds of it, measures them, sets the correction on the master, measures again with it (the bus compressor and the limiter come after the gain) and puts the rest right. That is some seconds of work: the piece begins at once, and the correction follows while it plays and glides in over a few seconds (the offline renderer measures before it renders). A concert's pieces each get their own, gliding from one to the next over eight bars. The correction is part of the piece: an export carries it from the first sample, and your master level comes on top. The loudest three seconds around -12 LUFS, the true peak at -1 dBTP, the peak-to-loudness ratio 11 to 14 dB and the loudness range 8 to 13 LU -- the dynamics are the product. eph_render prints these figures for everything it renders, with the stereo correlation and how far the side lies under the mid. Below, the composer's settings -- the style, key, scale, tempo, lengths and whether the style chooses the tempo -- which the top bar also shows.

Album Form makes a concert an album: an interlude of three to five minutes between the long pieces -- no rows, only the drone, a slow chord every 16 to 32 bars on the strings or the tape keys, the atmosphere with its grains, a longer hall and the open fifth at the end -- the piece where the concert's middle falls the darkest (Phrygian, darker voices, a row more at the peak), and the last one ethereal (Dorian, a longer hall, a row fewer, no drums, a quieter lead). Without it a concert is what it was.

Motion keeps the depth awake: a spectrum that does not change for twenty seconds loses its depth to the ear. The pads' levels move by a few tenths of a dB within a second or two; their hall sends, the strings' tone and the echo's feedback over 13 to 30 seconds; the atmosphere swells by 6 dB over 41 seconds -- periods that never lock into the sequence's cycles, so foreground and background do not breathe together. In the atmosphere, the bridges and the coda, where no sequence plays, a near event sounds now and then: a lead fragment of one to three notes, 20 to 90 seconds apart. Sub Solo leaves only what lies under 80 Hz, for checking the bottom of the mix; Mono and Sub Solo are for listening and change nothing that is exported unless they are on.

Exports (the plugin's and eph_render's) end with twenty seconds of the rooms after the last note, faded in over two seconds and out over the last ten, never linearly, so the last tail never breaks off. eph_render --archive writes an archive master besides: no limiter, 24 bit, the whole scaled to a true peak of -3 dBTP (it renders twice). The report adds the crest factor (true peak over RMS, 17 to 20 dB for the styles; 12 is the least that keeps the transients) and, with --stems, the correlation of each stem: the rows are mono points (1.0), the tape keys spread by all-pass decorrelation (about 0.35), the rooms enveloping (about 0.2).

OSC Cues sends what the score knows to a visualiser such as Kaleidoscope, over UDP at the OSC Port, to the machine named by the environment variable EPH_CUE_HOST (this machine without it): every beat with the tempo (/eph/beat), every section with its phase (/eph/phase), every new root (/eph/key) and every conjunction of the rows with the share of the running rows that meet (/eph/conjunction). Each cue leaves when its sound is heard, not when it is composed.

MASTER

Control	Key	Range	Default
Level	`master.level`	-60 .. 6 dB	0 dB
Compress	`master.compress`	0 .. 1	0.5
Ceiling	`master.ceiling`	-6 .. 0 dBTP	-1 dBTP
Width	`master.width`	-6 .. 6 dB	4.5 dB
Mono	`master.mono`	off / on	off
Cascade Duck	`master.cascade`	0 .. 6 dB	2.5 dB
Motion	`master.motion`	0 .. 1	1
Sub Solo	`master.sub_solo`	off / on	off
Soft Clip	`master.clip`	0 .. 2 dB	0.75 dB
Sub Ceiling	`master.sub_ceiling`	-18 .. 0 dBFS	-6 dBFS
Resonance Tamer	`master.tame`	0 .. 1	0.6

Export



What leaves the instrument, in one place, as every generator of the family has it: Export writes what plays as a 24-bit WAV and a MIDI file beside it, Export with stems adds a WAV per channel strip and one for the rooms, the disk and the folder save and load a piece as a small .ephset file (the seed, the lengths, the rerolls and every changed knob). The score cues tell a visualiser such as Kaleidoscope over OSC what the score knows, at the moment it is heard.

CUE

Control	Key	Range	Default
OSC Cues	`cue.enabled`	off / on	off
OSC Port	`cue.port`	1024 .. 65535	9000

Style



	Cosmic	Doom	Melodic	Modern	Drift	Custom
Tempo (BPM)	112 - 126	92 - 110	118 - 132	118 - 130	100 - 118	112 - 126
Piece in a concert (min)	12 - 22	10 - 18	7 - 12	6 - 11	12 - 28	12 - 22
Sequence phases	1 - 2	1 - 2	1 - 2	1 - 2	2 - 3	1 - 2
Atmosphere before / coda after	8 % / 10 %	9 % / 10 %	7 % / 8 %	6 % / 10 %	9 % / 12 %	8 % / 10 %
New tempo / new key in a phase	30 % / 30 %	20 % / 40 %	40 % / 40 %	50 % / 50 %	70 % / 60 %	30 % / 30 %
Rows at the peak	3	3	4	4	2	3
Counter row lengths	13 12 7 9	11 7 13	12 8 6 16	7 5 12 13	13 9	13 12 7 9
Mutation per cycle	20 %	15 %	10 %	30 %	30 %	20 %
Transposer	8 x 4 bars	6 x 4 bars	8 x 2 bars	8 x 2 bars	6 x 4 bars	8 x 4 bars
Tape keys	90 %	70 %	60 %	50 %	80 %	90 %
String machine	50 %	30 %	60 %	70 %	40 %	50 %
Lead section	60 %	30 %	90 %	60 %	50 %	60 %
Bleeps	40 %	20 %	20 %	60 %	50 %	40 %
Drums	0 %	5 %	80 %	50 %	0 %	0 %
Granular cloud	60 %	30 %	20 %	40 %	70 %	60 %
Tape set	Choir	Strings	Strings	Flute	Choir	Choir
Lead density	0.50	0.35	0.75	0.60	0.40	0.50
Moves: length / rest	8 s / 14 s	10 s / 16 s	7 s / 12 s	6 s / 11 s	12 s / 18 s	8 s / 14 s
Darkness	-0.05	-0.15	0.05	0.00	-0.08	-0.05
Hall	6.0 s	7.5 s	4.5 s	5.0 s	8.0 s	6.0 s
Level	4.8 dB	2.8 dB	3.3 dB	4.5 dB	4.0 dB	4.8 dB

The five style profiles side by side, the chosen one lit. A style is not a sound but a set of ranges and chances the composer draws from: the tempo, the length and the number of sequence phases, how much of the piece is atmosphere before the rows and after them, how often a later phase changes tempo or key, how many rows play at the peak and which lengths they are drawn from, how often a row mutates, how slowly the transposer moves, how often each layer comes in, how dense the lead is, how fast the hands move, how dark the voices are, how long the hall rings and how loud the piece is. Two pieces of one style share these, not their notes. The tempi and the levels were set after measuring reference recordings.

The sixth column is a style of your own. "Copy ... into Custom" takes the chosen style's numbers as a starting point; change what you like on the right, and turn on Use Custom Style: the composer then takes your numbers. The counter rows' lengths, the transposer, the tape set and the drum patterns still come from the chosen style -- Cosmic and Drift have no drum patterns, so there Drums Chance does nothing. Your style is saved with the set and the plugin's state, and a concert starts from it (and morphs from it).

CUSTOM

Control	Key	Range	Default
Use Custom Style	`custom.use`	off / on	off
Tempo Low	`custom.bpm_low`	60 .. 160 BPM	112 BPM
Tempo High	`custom.bpm_high`	60 .. 160 BPM	126 BPM
Piece Min Low	`custom.minutes_low`	4 .. 40 min	12 min
Piece Min High	`custom.minutes_high`	4 .. 40 min	22 min
Phases Low	`custom.phases_low`	1 .. 4	1
Phases High	`custom.phases_high`	1 .. 4	2
Atmosphere Share	`custom.intro`	0 .. 0.3	0.08
Coda Share	`custom.coda`	0 .. 0.3	0.1
New Tempo Chance	`custom.new_tempo`	0 .. 1	0.3
New Key Chance	`custom.new_key`	0 .. 1	0.3
Rows at Peak	`custom.peak_rows`	1 .. 7	3
Mutation	`custom.mutation`	0 .. 1	0.2
Tape Keys Chance	`custom.tape`	0 .. 1	0.9
Strings Chance	`custom.strings`	0 .. 1	0.5
Lead Chance	`custom.lead`	0 .. 1	0.6
Bleeps Chance	`custom.bleeps`	0 .. 1	0.4
Drums Chance	`custom.drums`	0 .. 1	0
Lead Density	`custom.lead_density`	0.1 .. 1	0.5
Move Length	`custom.hand_move`	2 .. 30 s	8 s
Rest Between Moves	`custom.hand_rest`	4 .. 40 s	14 s
Darkness	`custom.darkness`	-0.5 .. 0.5	-0.05
Hall	`custom.hall`	1 .. 15 s	6 s
Level	`custom.level`	-12 .. 12 dB	4.8 dB

With a DAW and other apps

Four things connect Ephemeris to the software around it and to the rest of the family (02.10.2026). The outputs, Link and the jam are off until they are wanted, the MIDI goes nowhere until a track takes it, and only the jam -- as a follower -- changes what the music sounds like.

MIDI out: in a DAW the plugin also sends what it plays -- every note of every part at the moment it sounds, a channel per part as in the MIDI file of an export -- and leaves the muted parts out. Record it on a MIDI track to keep a part as notes, or route it to another instrument to double it. A stop or a jump of the transport sends all notes off.

Outputs of their own: besides the main output the plugin has a stereo output for each of the fifteen channel strips -- the eight rows, the lead, the drone, the tape keys, the strings, the poly synth, the drums and the atmosphere -- and one for the rooms, the stems an export writes. They are off to begin with; switched on in the DAW (in most hosts: the plugin's output configuration, or its multi-output version), each carries its part before the master, while the main output plays on as before.

Ableton Link (the standalone; Settings > Ableton Link): with another Link app on the network -- Live, Bitwig, a phone app, another generator of the family -- Ephemeris takes the session's tempo, lines its bars up with the session's and starts and stops with it; alone in a session it offers its own tempo to whoever joins. In a DAW the host's transport leads, as before.

The family jam (Settings > Family jam, in the plugin and in the standalone; off to begin with): the five instruments of the family -- Totality, Parhelion, Ephemeris, Phosphene and Noctuary -- play as one band on the local network. One leads: its key, the energy of its sections, its breaks and its drops go out to the others. Those set to Follow take them up -- in Ephemeris: every voice but the drums moves to the leader's key, on top of its own Transpose; the rows' filters close with a quiet section of the leader's (while the performer's hand leaves them alone); and in the leader's breaks the drums are out until its drop. A new root comes in at the follower's next bar line, by the shortest way, the mode with its next piece. Tempo and bars come from Ableton Link in the standalone (switch it on in each of them) or from the DAW's transport, so that a section the leader announces lands on the same bar everywhere. As the leader Ephemeris says its piece's key and, a beat ahead, each phase -- the atmosphere, the entry, the build, the peak, the breakdown, the coda --, with its energy. The jam stays on this network (UDP multicast to 239.255.77.77, port 9177, not past the router), instruments on one machine find each other as well, and a follower that has not heard its leader for four seconds goes back to its own key and its own form. The last line of the submenu says what it does: off, leading, or whom it follows.

On a Mac: every release has a build for Apple Silicon (macOS 12 or newer), the standalone, the VST3 and the Audio Unit (Logic, GarageBand, MainStage; passed by Apple's auval), built and tested on GitHub's machines. It is signed ad hoc, not notarized: macOS asks once (README-macOS.txt in the zip says how), and it has not yet been played on a real Mac -- reports are welcome.

On Linux: every release has an archive for x86-64 (glibc 2.35 or newer: Ubuntu 22.04, Debian 12, Fedora 36 and later) with the standalone, the VST3, the LV2 (Ardour, Carla, Reaper, Qtractor) and the renderer, built and tested on GitHub's Ubuntu machines and tried under WSL. README-Linux.txt in it says where everything goes: the VST3 into ~/.vst3, the LV2 into ~/.lv2, the standalone started where it lies, sound through ALSA or JACK (PipeWire serves both).

Demos: the release "demos" holds a track per style as an MP3 and one of them as a video, with pictures by KaleidoscopeEnhanced cut to the track's own score cues; Tools/demo/make_demos.py makes them all again (eph_render).

KaleidoscopeEnhanced understands the score cues of all five instruments as they come (since 02.10.2026) -- Totality's blocks and keys, Parhelion's sections, Ephemeris' phases, Phosphene's sections and drops, Noctuary's bars, keys and scenes -- and cuts its pictures to them, on a drop at once.

Meta Quest

The whole generator runs on the headset (bin\quest\EphemerisQuest.apk, see Quest/README.md), played with the hands in the grammar every generator's headset shares (The panel): left pinch play and stop, right pinch the moves held and let go, both hands together the next piece; the left hand's height the rows' filters, the right's the echo throw. A head-locked panel shows the piece, its section, the root and the scale, the time and the two hand controls, the orrery of the Rack page beside it. A file eph.cfg in the app's folder sets the seed, the style, a piece's length, the quality and any knob; bridge_host (and bridge_port, 9102) send the hands to Ephemeris on a computer instead, which then plays with them as with its own; audio=0 leaves the headset silent.